

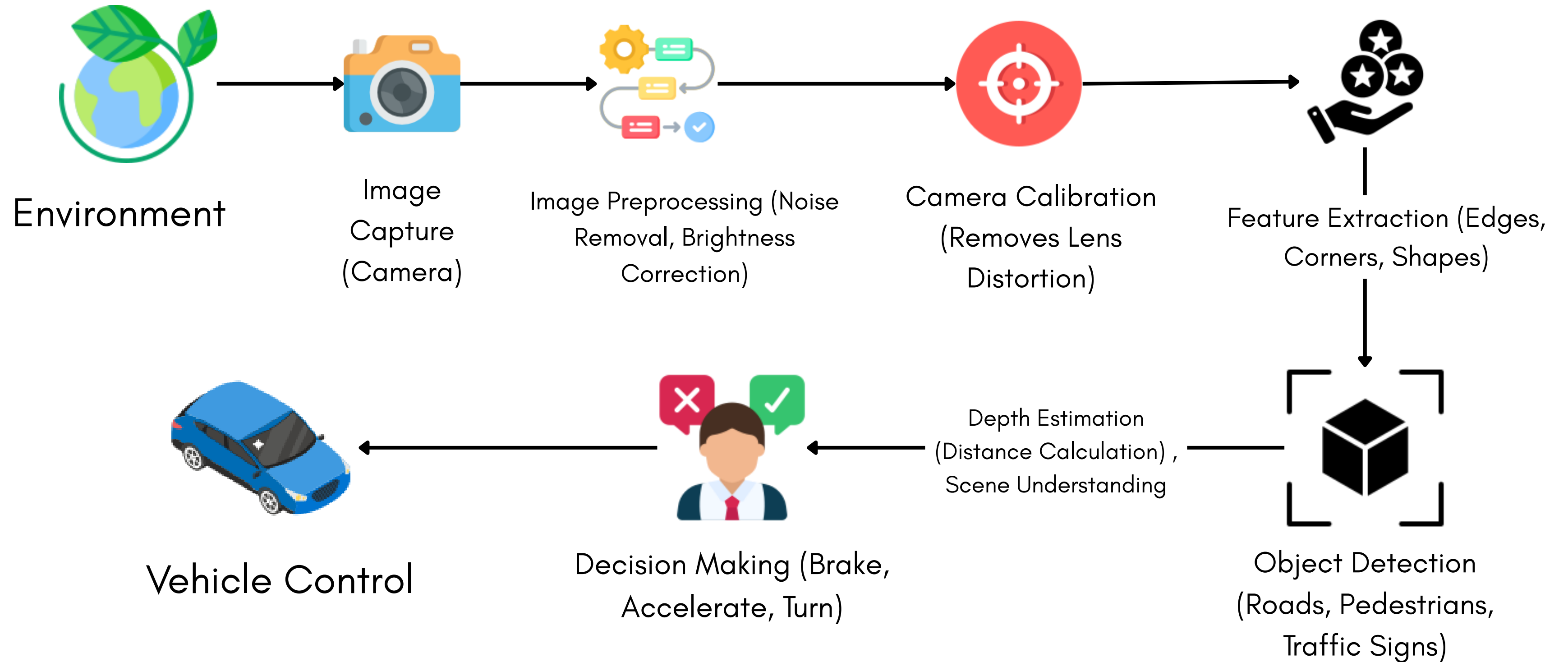
Technical Presentation

Autonomous Car Detects Roads, Pedestrians, and Traffic Signs

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Vision Pipeline



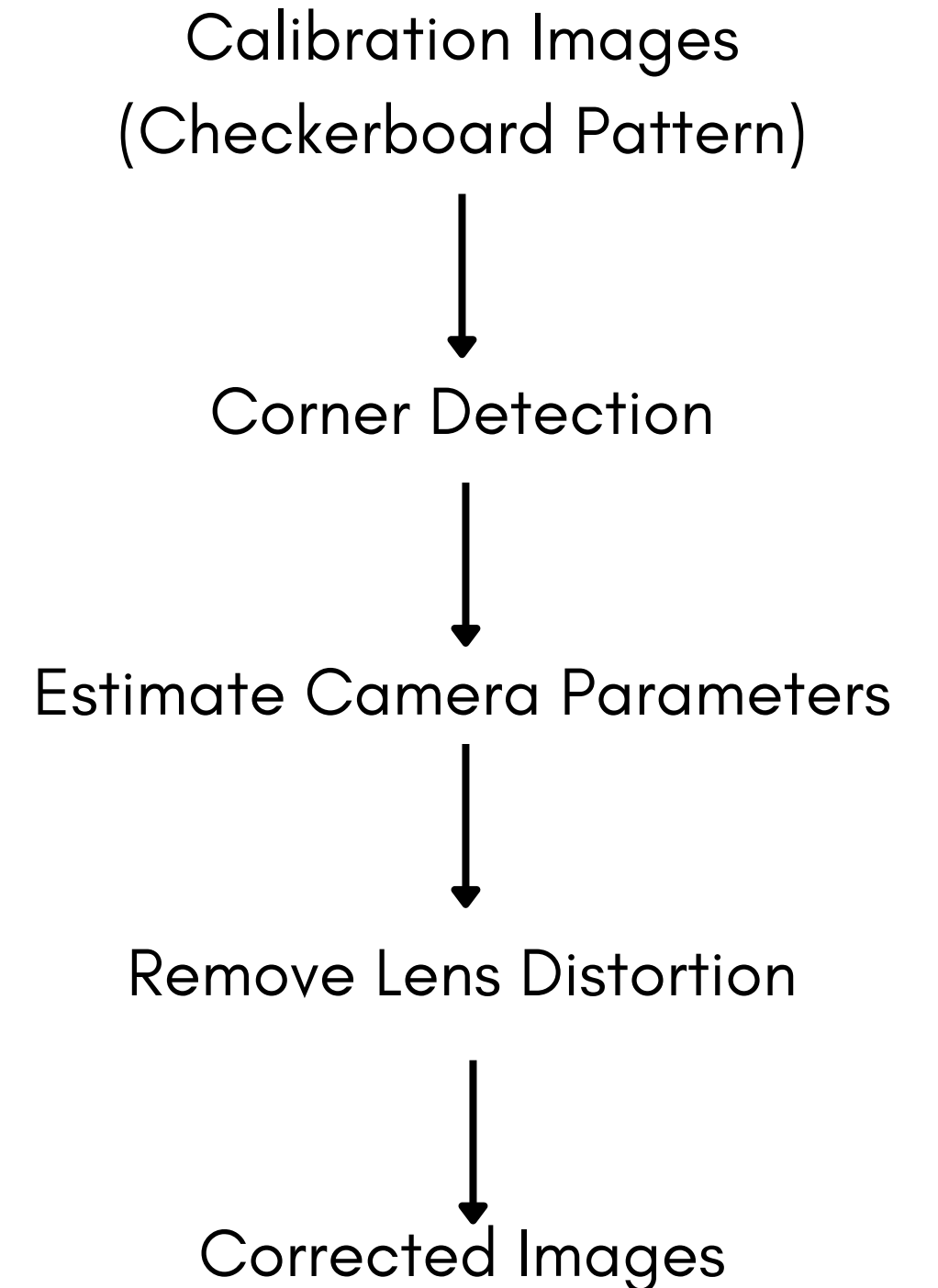
Camera Calibration

Definition:

Camera calibration determines the camera's internal and external parameters to produce accurate measurements.

Why Calibration is Needed?

- Removes lens distortion.
- Measures real-world distances accurately.
- Improves object detection.
- Essential for depth estimation.



Depth Estimation

Definition:

Depth estimation calculates the distance between the autonomous car and surrounding objects.

Why is Depth Important?

- Detects nearby obstacles and Maintains safe following distance.
- Prevents collisions.
- Supports parking and navigation.

1. Stereo Vision:

Uses two cameras like human eyes.

Nearby objects show larger disparity and are estimated to be closer.

2. Monocular Depth

Estimation:

Uses a single camera with deep learning models to estimate depth.

3. Sensor Fusion:

Combines:

- Camera
- LiDAR
- Radar

for more accurate distance estimation.

Conclusion

- The vision pipeline enables the car to capture, process, and interpret road scenes.
- Camera calibration ensures images are geometrically accurate by correcting lens distortion and estimating camera parameters.
- Depth estimation calculates the distance to roads, pedestrians, vehicles, and signs, enabling safe navigation.
- Together, these components allow autonomous vehicles to perceive their environment and make intelligent driving decisions.